FLEET TEAM DIGEST

737MAX-FTD-34-19012

Issue Title : Collins GLU-2100 MMR Temporary GPS Unavailability or Inaccurate GPS Position data at Northern Latitudes near 180 Degrees Longitude

Airplane Model	ATA	Minor Model(s)
737MAX	3458-00	-8, -9, -BBJ
Other Model(s)	**ECCN**	
737NG, 777	9E991	

Originated Date	Last Revised Date	Created On
12/23/2019	10/22/2020	10/16/2020
Estimated Completion Date	**Next Update Date**	

Revision Description

- Updated Final Action section to provide SB release information.

- Updated Milestones section to provide SB release date.

Applicability

This applies to 737-8 / -9 / BBJ airplanes delivered from production, or models that have been modified by Boeing Service Bulletin, which have the Collins Aerospace GLU-2100 MMR receiver installed.

Description

This FTD provides information regarding airplanes equipped with the GLU-2100 Multi-Mode Receiver (MMR) experiencing temporary GPS position data issues while operating at Northern latitudes near 180 degree longitude.

Background

Boeing has received reports from operators describing the loss of GPS data while using the Collins GLU-2100 MMR, PN: 822-2532-100, and flying routes at Northern latitudes near 180 degrees longitude. Reports indicated that the condition usually corrected itself, however MMR circuit breaker cycling was performed to recover MMR operation in some cases. The root cause has been identified as a miscalculation in the computation of the ionospheric corrections in Satellite Based Augmentation System (SBAS) mode while at the edge of coverage of the United States SBAS system (Wide Areas Augmentation System - WAAS), and can cause the Global Navigation Satellite System (GNSS) bus to shut down anywhere from a few seconds to up to 40 minutes or potentially output un-annunciated inaccurate GPS position data. Collins has indicated a MMR software update is necessary to correct this concern.

Status

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Collins Aerospace, the manufacturer of the unit, has determined the root cause of this event to be due to an improper implementation within OPS software (COL4E-0087-0001 or COL4D-0087-0002) which occurs in a funnel shaped area that mainly extends +/- 20 degrees on either side of 180 degrees W Longitude, and encompasses the Northern Hemisphere to 10 degrees Latitude in the Southern Hemisphere.

There are currently two known potential effects:

- An annunciated loss of GPS output, where the GNSS bus becomes inactive anywhere from a few seconds to up to 40 minutes. Other MMR functions (ILS, VOR, MB, etc.) are not affected and the GNSS bus will automatically recover once the aircraft leaves the region near the WAAS western edge of coverage.

- Un-annunciated reduced positional accuracy in the affected region when the SBAS ionosphere corrections are improperly applied (Positional error will be bounded to 0.3 nautical miles, but may not be bounded by the horizontal protection level (HPL) that is output by the GNSS).

Boeing has determined that the un-annunciated reduced positional accuracy in lateral deviation, during operations less than Required Navigation Performance (RNP) 0.3, can potentially affect continued safe flight and landing. Therefore, Boeing has expanded previous SRP: 737MAX-SRP-34-0165, related to the previous GLU-2100 week 9 anomaly, to encapsulate this concern as well.

Collins has resolved this condition in an updated operational software (OPS) that is currently anticipated to be released 2Q 2020. This OPS software update will resolve this improper implementation, along with a previous GPS Week 9 Anomaly matter (737MAX-FTD-34-19006).

The Federal Aviation Administration (FAA) has indicated that they will issue final regulatory action.The FAA has issued Airworthiness Directive (AD) 2020-03-20 as an Immediate Adoptive Rule (IAR) requiring the insertion of a copy on the AD into the applicable Airplane Flight Manual (AFM) (reference MOM-MOM-20-0128-01B).No timeline has been provided for final ruling.

Interim Action

Boeing recommends affected airplanes to be restricted to operations of RNP 0.3 or above.

Boeing recommends affected operators review Collins SIL, GLU-2100-19-3, for a map of the affected area and a list of affected airports. Additionally, Boeing is currently developing AFM updates to provide operators with further guidance and will issue a MOM message, upon its release.

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Final Action

Boeing has completed certification of the updated GLU-2100 MMR OPS software, PN: COL4C-0087-0003, and has released Alert Service Bulletin, 737-34A3572, to provide operators with retrofit instructions for software installation (On-Wing Loadable) into the GLU-2100.

Milestones

Note: Target dates are based upon currently available status and have risk of future revision.

Root Cause Established: 02-DEC-2019 (Complete)

SRP Solution Selection: 30-JAN-2020 (Complete)

SRP Plan Commit: 20-FEB-2020 (Complete)

Production Certification Complete: 25-MAR-2020 (Complete)

Production Incorporation: L/N 7973

Service Bulletin 737-34A3572 Available: 15-OCT-2020 (Complete)

Supplier Information

Rockwell Collins, Inc. Systems

400 Collins Rd. NE

Cedar Rapids, IA 52498-0001

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Service Information Letter (SIL)	Collins GLU-2100-19-3	2	02/20/2020
Service Related Problem (SRP)	737MAX-SRP-34-0165		
Multi-Operator Message (MOM)	MOM-MOM-19-0731-02B		
Fleet Team Digest Article (FTD)	737MAX-FTD-34-19006		
Multi-Operator Message (MOM)	MOM-MOM-20-0128-01B		

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Parts List

Part Name	Boeing Part Number	Supplier Name	Supplier Number
GLU-2100 MMR	822-2532-100	Collins Aerospace	

Related Categories

- Component Issues : Issues that occur during component maintenance or have an impact on component reliability
- Regulatory : Issues mandated by airworthiness directive or other regulatory guidance material
- SRPs - Safety : Service Related Problems in work by Boeing that have an airplane or personal safety impact

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